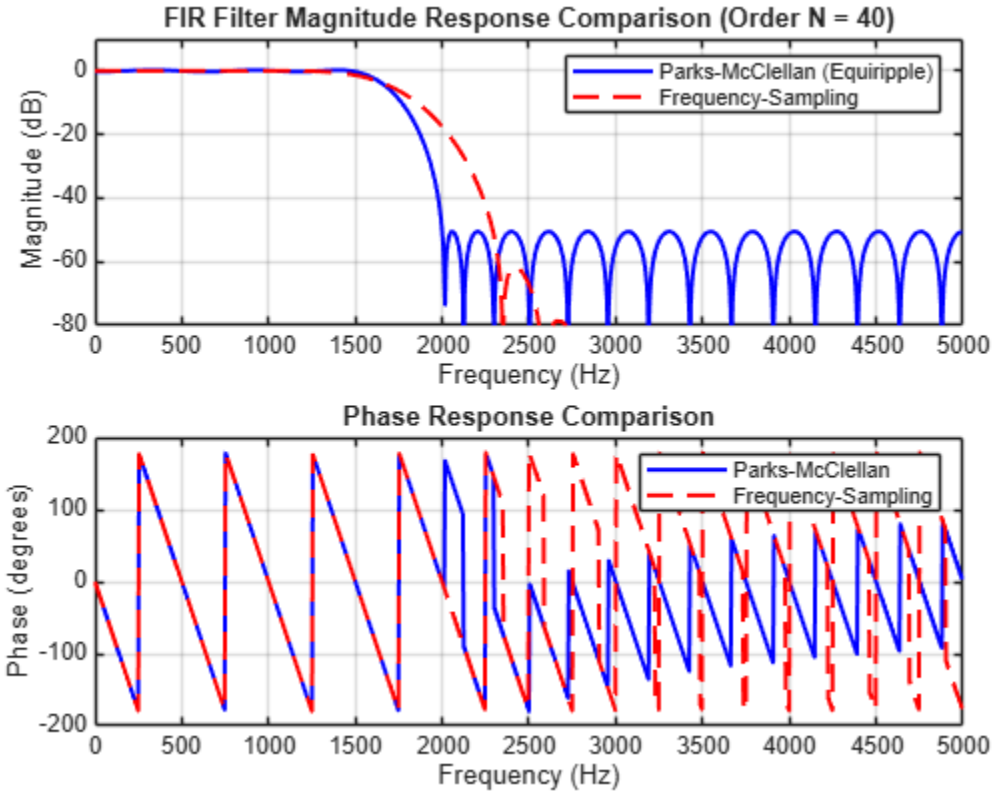
```
clear;
clc;
close all;

% ========================================
% EXPERIMENT 5:
% PARKS-MCCLELLAN vs FREQUENCY-SAMPLING
% ========================================

% Basic Parameters
Fs = 10000;       % Sampling frequency (Hz)
fp = 1500;        % Passband edge frequency (Hz)
fs = 2000;        % Stopband edge frequency (Hz)
N = 40;            % Filter order

% Filter specifications
Wp = 1;            % Passband weight
Ws = 10;           % Stopband weight
weights = [Wp Ws];

% Normalized frequency band edges
f_bands = [0 fp fs Fs/2] / (Fs/2);

% Desired amplitudes
a_bands = [1 1 0 0];

% ========================================
% 1. PARKS-MCCLELLAN (EQUIRIPPLE) DESIGN
% ========================================

b_pm = firpm(N, f_bands, a_bands, weights);

% ========================================
% 2. FREQUENCY-SAMPLING DESIGN
% ========================================

b_fs = fir2(N, f_bands, a_bands);

% ========================================
% DISPLAY NUMERICAL OUTPUTS
% ========================================

fprintf('========================================\n');
fprintf('EXPERIMENT 5: FIR FILTER DESIGN COMPARISON\n');
fprintf('========================================\n\n');

fprintf('Sampling Frequency: %d Hz\n', Fs);
fprintf('Filter Order (N): %d\n', N);
fprintf('Passband Edge: %d Hz\n', fp);
fprintf('Stopband Edge: %d Hz\n\n', fs);

fprintf('--- PARKS-MCCLELLAN COEFFICIENTS ---\n');
```

```
fprintf('First 10 coefficients:\n');
disp(b_pm(1:10));

fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS ---\n');
fprintf('First 10 coefficients:\n');
disp(b_fs(1:10));

% ========================================
% FREQUENCY RESPONSE
% ========================================

[H_pm, f] = freqz(b_pm, 1, 1024, Fs);
[H_fs, ~] = freqz(b_fs, 1, 1024, Fs);

% Magnitude response in dB
mag_pm = 20 * log10(abs(H_pm) + eps);
mag_fs = 20 * log10(abs(H_fs) + eps);

% Phase response in degrees
phase_pm = angle(H_pm) * 180 / pi;
phase_fs = angle(H_fs) * 180 / pi;

% ========================================
% PLOT COMPARISON
% ========================================

figure('Name', 'Parks-McClellan vs Frequency-Sampling');

% Magnitude Response
subplot(2,1,1);

plot(f, mag_pm, 'b', 'LineWidth', 1.5);
hold on;

plot(f, mag_fs, 'r--', 'LineWidth', 1.5);

grid on;

legend('Parks-McClellan (Equiripple)', ...
       'Frequency-Sampling');

title(['FIR Filter Magnitude Response Comparison (Order N = ', ...
       num2str(N), ')']);

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

ylim([-80 10]);
xlim([0 Fs/2]);

% Phase Response
subplot(2,1,2);

plot(f, phase_pm, 'b', 'LineWidth', 1.5);
```

```
hold on;

plot(f, phase_fs, 'r--', 'LineWidth', 1.5);

grid on;

legend('Parks-McClellan', 'Frequency-Sampling');

title('Phase Response Comparison');

xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

xlim([0 Fs/2]);

% ========================================
% END OF EXPERIMENT
% ========================================

==========================================
EXPERIMENT 5: FIR FILTER DESIGN COMPARISON
==========================================

Sampling Frequency: 10000 Hz
Filter Order (N): 40
Passband Edge: 1500 Hz
Stopband Edge: 2000 Hz

--- PARKS-MCCLELLAN COEFFICIENTS ---
First 10 coefficients:
  Columns 1 through 7

    0.0034    0.0030   -0.0016   -0.0090   -0.0121   -0.0055    0.0072

  Columns 8 through 10

    0.0136    0.0041   -0.0148

--- FREQUENCY-SAMPLING COEFFICIENTS ---
First 10 coefficients:
  Columns 1 through 7

    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008

  Columns 8 through 10

    0.0035    0.0029   -0.0039
```



Published with MATLAB® R2026b